MSBA 265 · APPLIED ANALYTICS · FINAL PROJECT · P1 SUBMISSION

Predicting Loan Payback

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1. Topic exploration: what each topic could do for a real organization

I considered all six topic areas. For each, one realistic example and the decision it could support:

Topic area	Realistic example	What the organization could learn or do
Supervised machine learning	A consumer lender predicts whether a borrower will repay a loan, using income, debt-to-income ratio, credit score, and loan terms.	Rank borrowers by risk, set interest rates, and send borderline files to a loan officer instead of reviewing every file by hand.
Unsupervised machine learning	A credit card issuer clusters customers by spending, payment timing, and utilization; a separate model flags transactions that look unlike a customer's usual pattern.	Design different credit-limit and collection strategies for each segment; send unusual transactions to a fraud review queue.
Dimensionality reduction	A bank's risk team applies principal component analysis (PCA) to dozens of correlated financial ratios for its small-business borrowers.	Monitor portfolio health with three or four summary factors on one dashboard, while documenting how much variation each factor keeps or loses.
Natural language processing	A bank classifies the free-text narratives in customer complaints by product and issue (for example, "billing dispute" or "unauthorized charge").	Route complaints to the right team faster and count which problems recur each month.
Semantic or sentiment analysis	A credit union scores the tone of public app-store reviews of its mobile banking app before and after an update.	See whether an update made customers happier or angrier and which features to fix first, while checking sarcasm and mixed reviews by hand.
Social media / digital channel analytics	A community bank compares engagement on its own public posts about budgeting, scams, and first-home buying, using the platform's export of its page data.	Decide which financial-education topics and posting times reach the most people, without collecting private messages.

Credit scoring: from expert judgment to explainable machine learning

Supervised classification has been used to predict loan repayment for nearly 70 years. What changed is the model, and what never changed is the duty to explain.

Topic: supervised ML
Industry: consumer lending

ORIGINS AND DEVELOPMENT

Scoring industry · Machine learning · Regulation

1956
Fair, Isaac and Company, known today as FICO, is founded [14].

Late 1950s
Credit scoring first emerges to support credit decisions at large retail stores and finance companies; by the end of the 1970s, most of the largest US commercial banks use scoring systems [1].

1974
The Equal Credit Opportunity Act (ECOA). Under Regulation B, reasons given for an adverse action "must be specific and indicate the principal reason(s)" [2].

1989
FICO and the three credit bureaus release the FICO Score. Generic, bureau-based scores that work across many loan products appear at the end of the 1980s [1] [14].

2004
Basel II lets banks use their own internal rating models to estimate the probability of default for capital rules [3].

2016–2017
XGBoost and LightGBM make gradient-boosted trees fast and accurate on large tables [4][5]. SHAP explains each prediction feature by feature [6].

2021
FinRegLab: machine learning underwriting is "intensifying" in US credit markets, most advanced in credit cards and unsecured personal loans [7].

2022
CFPB Circular 2022-03: using a complex algorithm does not excuse a lender from giving specific reasons [8]. (Withdrawn in May 2025; the Regulation B requirement itself remains [2][8].)

2024
Bank of England and FCA survey: 75% of UK financial firms already use AI; gradient boosting is the most common model type [9].

PRACTICAL USE TODAY

75% of UK financial firms use AI (58% in 2022) [9]

32% of AI use cases are gradient boosting models [9]

64% use SHAP to explain AI models (72% use feature importance) [9]

WHO USES IT, AND FOR WHAT

- **Lenders using credit scores:** to approve applications, set interest rates by risk, manage accounts, and estimate losses [1].
- **Banks:** scorecards for approval; internal rating models to estimate default risk for capital [3].
- **Card and personal-loan lenders, fintechs:** ML underwriting that evaluates tens of thousands of applicants each week [7].
- Because reasons must be specific [2], ML is often tested as a **challenger** next to a simpler, explainable scorecard.

VALUE AND IMPACT

POSSIBLE BENEFITS

Better risk ranking can mean fewer defaults at the same loan volume, or more approvals at the same loss; faster and more consistent decisions [7].

COSTS AND RISKS

Lenders must still give specific reasons [2]; models must be tested for unfair impact on protected groups; complex models are harder to validate and can drift.

Benefits depend on the lender and data; they are not guaranteed savings.

CONNECTION TO MY PROJECT

- **Champion vs. challenger:** the data includes the lender's own grade (A1 to F5), so I can test ML against the lender's current ranking, like a real model review.
- **Method:** logistic regression (transparent) and LightGBM (the industry's most common model type [9]).
- **Explainable:** feature contributions (the same idea as SHAP [6]) give each borrower's main risk drivers.
- **Early evidence:** AUC 0.922 vs. 0.655 for the lender's grade (synthetic data; see Section 5).

[1] Federal Reserve Board (2007), Report to the Congress on credit scoring.
[2] CFPB, Regulation B, 12 CFR §1002.9.
[3] Basel Committee on Banking Supervision (2004), Basel II revised framework.
[4] Chen & Guestrin (2016), XGBoost, KDD.
[5] Ke et al. (2017), LightGBM, NeurIPS.
[6] Lundberg & Lee (2017), SHAP, NeurIPS.
[7] FinRegLab (2021), ML for credit underwriting: market context.
[8] CFPB (2022), Circular 2022-03; withdrawal: Morgan Lewis CFPB guidance tracker.
[9] Bank of England & FCA (Nov. 21, 2024), AI in UK financial services.
[14] myFICO (2018), The history of the FICO Score.

Full links in the reference list at the end of this document.

3. Selected focus and project proposal

Selected focus: supervised machine learning (binary classification)

Element	My project
Problem	Lenders already rank borrowers with their own risk grade, but about 1 in 5 approved loans in this data is not repaid. A better ranking would leave less money in loans that default.
Intended user	A consumer lender's credit risk team: the credit risk manager who sets policy, and loan officers who review borderline files.
Data source	Kaggle Playground Series S5E11, "Predicting Loan Payback" [10]: 593,994 labeled loans for training and 254,569 unlabeled loans for scoring.
Method	Supervised binary classification: logistic regression (course method, explainable) and LightGBM gradient boosting, compared with a majority-class baseline, the lender's own grade, and a classic credit score + DTI scorecard.
Action supported	Group similar future applicants into low, medium, and high risk tiers (approve quickly / officer review / decline), and show the main drivers of each borrower's default risk.

Business problem and stakeholder

A consumer lender has already approved and funded many personal loans, and about 20% of them were not repaid. The lender ranks borrowers with its own risk grade (A1 to F5). The **credit risk manager** wants to know whether a machine learning model ranks borrowers by default risk well enough to be worth adopting, and **loan officers** need to understand why a borrower is flagged as risky.

> **Project question.** Can machine learning rank approved borrowers by default risk better than the lender's own grade, by enough to matter, while staying explainable?

Intended use

- **How much better, in dollars:** if the lender kept only its safest 70%, 80%, or 90% of loans, how much defaulted principal would be left under each ranking method?
- **What drives the risk:** the top three drivers of each high-risk borrower's score (model explanations, not real decline reasons).
- **How to use it:** low / medium / high risk tiers for similar future applicants (approve quickly / officer review / decline), with cutoffs set by stated cost assumptions and a sensitivity check. This mirrors the PASS / REVIEW / REJECT decision in our classification lecture [11].

Success measures

Measure	What counts as useful
ROC AUC, 5-fold cross-validation	Clearly above the lender's grade (0.655) and a classic credit score + DTI scorecard (0.768).
Defaulted principal kept	Less defaulted principal than the lender's grade at every share of loans kept (70 / 80 / 90%).
Recall on loans not repaid; Brier score	Catches more defaults than the baselines; predicted probabilities are accurate, not just well ranked (lower Brier score).
Explainability	Each high-risk borrower gets readable top drivers that agree with the EDA.

Dataset and access

- **Source and access:** Kaggle Playground Series S5E11 [10]. Readers sign in to Kaggle, accept the competition rules, and download the files into `data/` (steps in the README). The competition rules do not allow redistributing the data, so no data files are stored in the public repository.
- **One row** = one loan that was already approved: the borrower's finances, loan terms, and profile, plus whether the loan was paid back.
- **Target:** `loan_paid_back` (1 = repaid, 0 = not repaid).

Key field	Business meaning	Expected risk signal
employment_status	Current job status: stability of income	Unemployed or student → much higher risk
debt_to_income_ratio	Existing debt payments ÷ income	Higher DTI → higher risk
credit_score	Bureau score (395 to 849 in the data)	Higher score → lower risk
grade_subgrade	Lender's own risk grade, A1 (best) to F5 (worst)	The lender's current ranking: the "champion"
annual_income, loan_amount, interest_rate	Ability to repay, exposure, and the price charged for risk	Used as numeric features
marital_status, education_level, loan_purpose	Borrower profile and use of funds	Weak signals (one-hot encoded)
gender	Borrower gender	Excluded from models; used only for a fairness check

Limits. (1) **Synthetic data:** generated by a deep learning model trained on an original loan dataset, so dollar results show direction, not real-bank amounts. (2) **Approved loans only:** declined applicants are not observed (the reject-inference problem), so results describe ranking approved loans. (3) **Privacy:** no names or personal identifiers; `id` is a row number. (4) **Licensing:** used under the Kaggle competition rules, which prohibit sharing the competition data with non-participants; the repository shares code, figures, and results only.

Planned method and why it fits

- **Task:** supervised binary classification, because the target is a 0/1 label and the output needed is a probability of payback [11].
- **Shortlist for tables:** "logistic + tree ensemble" is the course's recommended starting shortlist for tabular data [11].
  - **Logistic regression**: the transparent first challenger. It turns a linear score $z = b + w \cdot x$ into a probability with the sigmoid, and its coefficients are fitted by maximum likelihood with gradient-based optimization [11][12].
  - **LightGBM** (gradient boosting): sequential trees that can learn the nonlinear DTI threshold found in the EDA [5][11].
- **Baselines:** always predict "repaid"; the lender's grade alone; a classic credit score + DTI scorecard.
- **Evaluation:** preprocessing fitted inside a pipeline to avoid leakage; stratified 5-fold cross-validation; ROC AUC (Kaggle metric), recall on loans not repaid, Brier score (ranking is not calibration); a held-out 20% validation set for the dollar analysis and tier cutoffs; Kaggle's test set scored once [11].

4. Initial data exploration

Data profile

Check	Result	Preparation decision
Size	593,994 training rows × 13 columns (11 features, id, target); 254,569 test rows	Large enough for a fixed 20% validation set (≈119,000 rows)
Missing values and duplicates	None in train or test	No imputation needed
Target balance	79.9% repaid, 20.1% not repaid	Accuracy is misleading; use ROC AUC and recall
Category consistency	No spelling or case clashes; train and test have the same levels	Keep "Other" as a real category
Outliers (Tukey 1.5 × IQR)	About 3% of income and DTI values flagged	Keep them: high-DTI outliers repaid only 45% vs. 81% for others, so they are real high-risk borrowers
Skewness	Income and DTI are right-skewed	Log transform for logistic regression
Correlation	Credit score and grade: r = −0.91	Logistic regression keeps credit score only

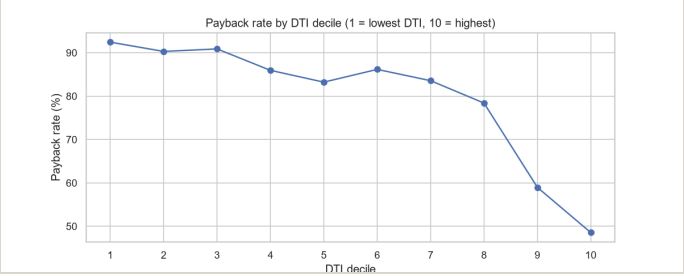


Figure 1. Payback rate by DTI decile (training data).

Interpretation. Payback stays between about 83% and 92% for the first seven deciles, then drops to 78%, 59%, and 49% in the top three. Risk rises sharply once DTI is high, which is a nonlinear pattern that tree models can learn directly.

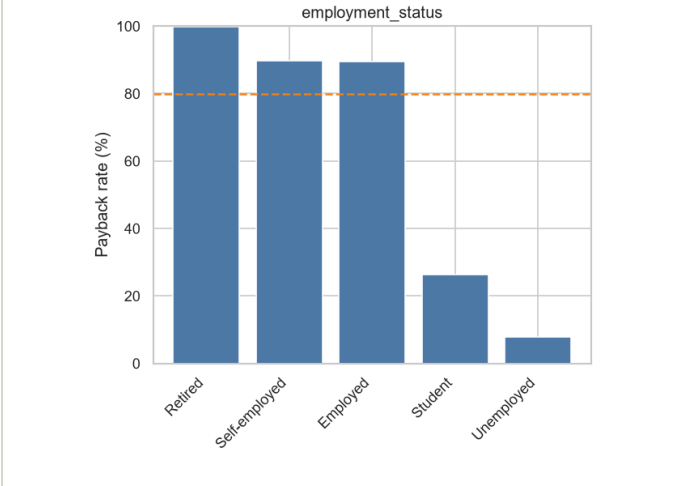


Figure 2. Payback rate by employment status (dashed line = overall 79.9%).

Interpretation. Employment status is the strongest single signal: only 7.8% of unemployed borrowers repaid, versus 89.4% of employed borrowers. Students are also far below average. Marital status, education, gender, and loan purpose barely move the payback rate.

What the EDA means for the model plan. The data is clean, so the work is in representation rather than repair: log-transform skewed numbers, one-hot encode small categories, encode the 30 grades in order (A1 = 1 … F5 = 30), exclude gender, and fit all preparation on training rows only. The strong employment and DTI signals suggest the project question is feasible.

Full audit and charts: `01_eda.ipynb` and `src/eda_audit.py`; tables and figures in `reports/`.

5. Baseline and proof of concept

I ran the full comparison as a feasibility test (`02_modeling.ipynb`, about 7 minutes to run). Scores are means across stratified 5-fold cross-validation on the training data.

Model	ROC AUC	Recall (not repaid)	Brier score
Baseline: always "repaid"	0.500	0.0%	0.201
Lender's grade only (company baseline)	0.655	1.8%	0.154
Classic scorecard: credit score + DTI	0.768	19.3%	0.135
Logistic regression	0.908	59.8%	0.077
LightGBM	**0.922**	**60.9%**	**0.072**

Recall uses the default 0.5 threshold; the risk tiers below use business-based cutoffs instead.

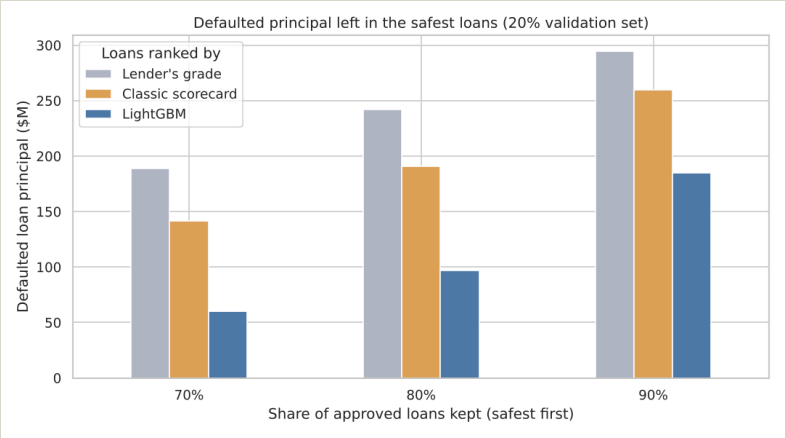


Figure 3. Defaulted principal left when only the safest share of approved loans is kept (20% held-out validation set, 118,799 loans).

Interpretation. Both models beat every baseline. Keeping the safest 80% of approved loans, ranking by LightGBM leaves **$97.1M** of defaulted principal, versus **$242.0M** with the lender's grade and $190.6M with the classic scorecard: about **60% less** than the grade. The gap holds at 70% and 90%.

This is principal on synthetic data, not final loss, so the direction matters more than the exact dollars.

First look at the two business outputs

Output	First result (illustrative)
Main drivers of high default risk	Among the riskiest 10% of validation borrowers (97.6% actually defaulted), the top driver is employment status for all of them; typical top three: unemployed, high DTI, low credit score.
Low / medium / high risk tiers	With a 60% loss given default, $50 per review, and review capacity of 15%: low risk ≥ 0.89 payback probability, medium 0.77–0.89, high < 0.77. Default rates: 3.7% / 15.4% / 67.1%. Cutoffs move with the assumptions, but net value stays positive in all nine scenarios tested.

> **Feasibility verdict.** The data supports the question, the models run in minutes, and the early results already meet the AUC and dollar success measures. The remaining work is about making the result trustworthy and reusable, not about whether it works.

6. Open questions and next steps

What remains unknown

- **Transfer to real data:** will a model trained on synthetic rows hold up on the original loan dataset?
- **Cost assumptions:** loss given default, review cost, and officer capacity are assumptions, not facts from the data.
- **Fairness:** do error rates differ across gender groups even though gender is not a model input?
- **Calibration:** a strong AUC does not prove that "80%" means 80% [11]; a calibration check is still needed.

Plan for P2

Step	What I will do	Course link
Tune and finalize	Tune LightGBM with cross-validation only (never on the Kaggle leaderboard); report the confusion matrix, precision, and recall at the tier cutoffs.	Classification: slides 7, 29–31 [11]
Calibration and fairness	Calibration curve and Brier score; compare error rates across gender groups.	Slide 32 [11]
Extension	Test the model on the original loan dataset to see how much AUC changes.	Honest evaluation [11]
Model handoff (production-readiness feature)	Save the fitted pipeline with joblib, plus a metadata file (feature order, class meaning, tier cutoffs, library versions, a reference prediction). Reload it in a fresh session and show it reproduces the reference prediction.	Module 3: model handoff [13]
Local API	A FastAPI `/predict` endpoint that takes one applicant, validates inputs (returns 422 for missing or invalid fields), and returns payback probability, risk tier, and top drivers; packaged in Docker.	Module 3: FastAPI and Docker lab [13]
Communicate	Data-to-action pipeline visual, draft paper, stakeholder PowerPoint, and a peer reproduction from the README.	Final project Part 5

References

[1] Board of Governors of the Federal Reserve System. (2007, August). *Report to the Congress on credit scoring and its effects on the availability and affordability of credit*. https://www.federalreserve.gov/boarddocs/RptCongress/creditscore/general.htm

[2] Consumer Financial Protection Bureau. (n.d.). Regulation B (Equal Credit Opportunity Act), 12 CFR §1002.9: Notifications. https://www.consumerfinance.gov/rules-policy/regulations/1002/9/

[3] Basel Committee on Banking Supervision. (2004, June). Basel II: International convergence of capital measurement and capital standards: A revised framework. Bank for International Settlements. https://www.bis.org/publ/bcbs107.htm

[4] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD Conference*. https://arxiv.org/abs/1603.02754

[5] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems 30*. https://proceedings.neurips.cc/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf

[6] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems 30*. https://arxiv.org/abs/1705.07874

[7] FinRegLab. (2021, September 16). The use of machine learning for credit underwriting: Market & data science context. finreglab.org (research report PDF)

[8] Consumer Financial Protection Bureau. (2022). Consumer Financial Protection Circular 2022-03: Adverse action notification requirements in connection with credit decisions based on complex algorithms. consumerfinance.gov. Withdrawn May 12, 2025: Morgan Lewis, CFPB guidance tracker: https://www.morganlewis.com/topics/cfpb-guidance-tracker-rescinded-and-remaining

[9] Bank of England & Financial Conduct Authority. (2024, November 21). Artificial intelligence in UK financial services – 2024. https://www.bankofengland.co.uk/report/2024/artificial-intelligence-in-uk-financial-services-2024

[10] Kaggle. (2025). Predicting Loan Payback: Playground Series, Season 5, Episode 11. https://www.kaggle.com/competitions/playground-series-s5e11
[11] MSBA 265. (2026). Classification foundations lecture (slides 7–9, 11–12, 20, 27–32, 34). University of the Pacific course materials.
[12] MSBA 265. (2026). Logistic regression lecture and Likelihood / MLE worksheet. University of the Pacific course materials.
[13] MSBA 265. (2026). Module 3 lab guide: From trained model to local API: FastAPI & Docker containerization. University of the Pacific course materials.
[14] Kaufman, R. (2018, August 21). The history of the FICO® Score. myFICO. https://www.myfico.com/credit-education/blog/history-of-the-fico-score